# One Carrier, Two Readouts: Born, Lorentz, Bloch, and Killing

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One-sheet companion to *On Information Physics: From State Distinguishability to Causal Accumulation*

**Origin.** CAL is not a theory of quantum gravity. It is an information-physics engine for non-Markovian reality, built to lift self-attention to a field-theoretic form from first principles of cognition, causality and intelligence. Its derivations of known physics are unit tests: a law of causal memory that cannot recover established physics has no business modeling minds or societies.

**Definition 1 (Information).** Information := distinguishability of physical states.

**Remark 1 (Difference and observation).** To distinguish physical states is to observe a difference between them. Change is a difference between states along an ordered evolution; thus change is temporally ordered distinguishability. Observation here denotes physical distinguishability, not a required conscious observer.

**Definition 2 (Causality).** Causality := change in physical state, including its rate of change.

**The law.** $T(x) = \alpha J(x) - (1-\alpha)\int_{J^-(x)} k(x,x')\,U(x,x')\,J(x')\,d^4x'$, with $J = \sum_a Z_a e_a$, $Z_a = A_a e^{-H_a/\tau_E} + B_a e^{iS_a/\hbar_{\text{eff}}}$, $A_a + B_a = 1$; $k$ is the memory kernel and $U$ the spacetime and internal transports (On Information Physics, Eq. 61, scalar-abbreviated).

**Definitions and symbols**

| Sym. | Meaning | Type | Eq. |
|---|---|---|---|
| $K(t,\tau),\ F$ | Memory kernel; aspect of past state | kernel; map | (5) |
| $M(t)$ | History functional | path functional | (5) |
| $\tau_E,\ \hbar_{\text{eff}}$ | Energy scale; action scale | positive reals | (9) |
| $e_a$ | $e_0 = I_2,\ e_j = -i\sigma_j$ | $M_2(\mathbb{C})$ | (10) |
| $J$ | Carrier $\sum_a Z_a e_a$ | $\mathbb{C}\otimes\mathbb{H}\cong M_2(\mathbb{C})$ | (11) |
| $\varrho$ | Positive closure $JJ^\dagger$ | Hermitian, $\succeq 0$ | (12) |
| $E_m,\ p_m$ | Positive effects; Born probabilities | $\succeq 0$; $[0,1]$ | (13) |
| $x^\mu,\ \eta_{\mu\nu}$ | Closure four-vector; Minkowski metric | real | (16) |

| Sym. | Meaning | Type | Eq. |
|---|---|---|---|
| $\alpha \equiv \alpha_{\text{frac}}$ | Interaction depth; fractional order | real | (19) |
| $I^\alpha$ | Past-supported Weyl integral | operator | (19) |
| $\theta$ | Accumulated Wick angle $\pi\alpha/2$ | angle | (21) |
| $\hat\varrho,\ \vec r$ | Normalized closure; Bloch vector | state; $\lvert\vec r\rvert \le 1$ | (29) |
| $G$ | $SL(2,\mathbb{C})$ element | Lie group | (36) |
| $\kappa$ | Killing form $\mathrm{tr}(\mathrm{ad}_A\mathrm{ad}_B)$ | bilinear form | (49) |
| $T^{\mu\nu},\ w_i$ | Dyad sum; dyad weights | tensor; real | (57) |
| $W,\ \pi^\mu$ | Action weight; canonical momentum | positive fn.; field | (62) |

### 1. History is more than its endpoints

*The fundamental theorem of calculus recovers net change. A history functional can tell apart two trajectories with the same endpoints.*

$$X(t) - X(t_0) = \int_{t_0}^{t} \dot X\, d\tau, \qquad M(t) = \int_{t_0}^{t} K(t,\tau)\,F\big(X(\tau)\big)\,d\tau.$$

*Only two kernels make the present sufficient: the Dirac limit (forgetful) and the light-cone limit (faithful). Because $K$ depends on the evaluation time, $M$ is not additive under concatenation, so it is not a functor to any additive group; its carrier is the path category $\mathrm{Path}(S)$, outside the de Rham tiers.*

### 2. Two weights in one channel

*Each channel holds a Boltzmann weight and a Feynman weight. The substitution $1/k_BT \to 1/(i\hbar)$ turns one into the other, the move that makes the path integral a partition function with quantropy as its entropy.*

$$Z_a = A_a e^{-H_a/\tau_E} + B_a e^{iS_a/\hbar_{\text{eff}}}, \qquad J = \textstyle\sum_{a=0}^{3} Z_a e_a \in M_2(\mathbb{C}).$$

### 3. The Born rule from positive closure

*Closing the carrier with its own adjoint gives a positive matrix. Its trace adds up the channels; positive effects read probabilities off it.*

$$\varrho = JJ^\dagger \succeq 0, \quad \tfrac{1}{2}\operatorname{tr}\varrho = \textstyle\sum_a |Z_a|^2, \quad p_m = \dfrac{\operatorname{tr}(E_m JJ^\dagger)}{\operatorname{tr}(JJ^\dagger)}.$$

*For $J = \psi\eta^\dagger$ and $E_m = \phi_m\phi_m^\dagger$ this is $p_m = |\phi_m^\dagger\psi|^2/\psi^\dagger\psi$. The inputs are positive effects and trace evaluation; the squared amplitude is algebraic.*

### 4. Lorentz from the determinant

*The same matrix has a real Hermitian slice. Its determinant is the Minkowski interval, and $SL(2,\mathbb{C})$ congruence preserves it.*

$$X = x^\mu\sigma_\mu, \quad \det X = (x^0)^2 - |\vec x|^2, \quad \det(GXG^\dagger) = \det X.$$

*The closure sits in the future cone: $\det\varrho = |\det J|^2 \ge 0$, rank one is null, rank two is timelike. This is the $\mathbb{C}$ case of $\mathfrak{sl}(2,\mathbb{K}) \cong \mathfrak{so}(\dim\mathbb{K}+1,1)$.*

### 5. The Bloch ball is a slice of the cone

*Normalize the trace to one and the future cone's cross-section is the Bloch ball.*

$$\hat\varrho = \frac{\varrho}{\operatorname{tr}\varrho} = \tfrac{1}{2}\big(I_2 + \vec r\cdot\vec\sigma\big), \quad |\vec r| \le 1, \quad \det\hat\varrho = \tfrac{1}{4}\big(1-|\vec r|^2\big).$$

*Surface: pure, null closures. Interior: mixed, timelike closures. Rotations keep the normalization; boosts do not, so the Bloch section is a normalization, not a Lorentz-invariant slice.* ***[Derived]*** *$2\sqrt{\det\varrho}/\operatorname{tr}\varrho = \sqrt{1-|\vec r|^2} = 1/\gamma = \sqrt{2S_L}$, the concurrence of a purification ($S_L = 1 - \operatorname{tr}\hat\varrho^2$); pure closures are null.*

### 6. The Killing form on the generator slice

*On traceless anti-Hermitian generators the determinant is the Killing form of $\mathfrak{su}(2)$, up to scale.*

$$\det A = |\vec y|^2 = -\tfrac{1}{2}\operatorname{tr}A^2 = -\tfrac{1}{8}\,\kappa(A,A), \qquad A = y^j e_j.$$

*On all of $\mathfrak{u}(2)$ the Killing form misses the centre; the determinant restores it with opposite sign, $\det A = -(y^0)^2 + |\vec y|^2$. Trace gives probability, determinant gives the metric, Killing gives the internal symmetry: three readings of one carrier.*

### 7. Real slices of one complex form

*Over $\mathbb{C}$, $\det J = Z_0^2 + Z_1^2 + Z_2^2 + Z_3^2$ has no signature. Choosing which channels carry an $i$ chooses the slice.*

| | | |
|---|---|---|
| $\mathbb{H}$ | $(+,+,+,+)$ | quaternion norm |
| $i\mathfrak{u}(2)$ (Hermitian) | $(+,-,-,-)$ | closures, Born, Minkowski |
| $\mathfrak{u}(2)$ (anti-Hermitian) | $(-,+,+,+)$ | transport generators |

*Minkowski is the Euclidean form with the three vector channels rotated by $i$: the algebraic quarter turn of Wick rotation.* ***[Derived]*** *Multiplication by $i$ swaps the state and generator slices isometrically. The two pairings use two bars: Born uses $^\dagger$; the Lorentz pairing applies the bar $X \mapsto \mathrm{adj}\,X = \epsilon X^T\epsilon^T$ ($\epsilon = i\sigma_2$) once, $X\,\mathrm{adj}\,X = \det X\, I_2$; applying it twice returns $X$, giving back the Born pairing on the Hermitian slice.*

### 8. Order is kept, but lossily

*Left-acting histories are remembered by $JJ^\dagger$ and erased by $J^\dagger J$.*

$$J_{tx}J_{tx}^\dagger = U_tU_x\varrho U_x^\dagger U_t^\dagger \;\neq\; J_{xt}J_{xt}^\dagger, \qquad J_{tx}^\dagger J_{tx} = J_{xt}^\dagger J_{xt} = J^\dagger J.$$

*The closure distinguishes some histories and identifies others: neither faithful nor forgetful. The commutator of two closures is anti-Hermitian, $[\varrho_1,\varrho_2]^\dagger = -[\varrho_1,\varrho_2]$* ***[Derived]****, so order-sensitive content lives on the generator slice; the Jarlskog CP invariant $\det[H_u, H_d]$ is of this form* ***[Proposed]****.*

### 9. Fractional accumulation keeps both readouts

*The Weyl integral rescales the decay sector and rotates the phase sector. The rotation between them is the Wick angle.*

$$I^\alpha e^{st} = s^{-\alpha}e^{st}, \quad I^\alpha Z_a = A_a\lambda_a^{-\alpha}e^{\lambda_a t} + B_a(i\omega_a)^{-\alpha}e^{i\omega_a t}, \quad \theta = \frac{\pi\alpha}{2}.$$

*$\theta$ is a quarter turn at $\alpha = 1$ and a half turn at $\alpha = 2$. Since $\varrho^{(\alpha)} = (I^\alpha J)(I^\alpha J)^\dagger \succeq 0$, the Born and cone identities hold at every order.*

### 10. Energy conditions from positivity

*Build a rank-two tensor from closure vectors. Positive weights on future-causal vectors give the energy conditions with no extra input.*

$$T^{\mu\nu} = \textstyle\sum_i w_i x_i^\mu x_i^\nu, \qquad T_{\mu\nu}u^\mu u^\nu = \sum_i w_i (x_i\cdot u)^2 \ge 0,$$
$$T^\mu{}_\nu u^\nu = \textstyle\sum_i w_i (x_i\cdot u)\,x_i^\mu, \qquad T^\mu{}_\mu = \sum_i w_i \det\varrho_i.$$

*In the accumulation the weights are $\alpha_{\text{frac}}$ (present) and $(\alpha_{\text{frac}}-1)k$ (history). With $k \ge 0$: $\alpha_{\text{frac}} \ge 1$ gives weak, null, and dominant energy conditions automatically.*

### 11. A weight gradient acts as friction [Derived]

*Weight the action by a positive function and the Euler–Lagrange equations gain a term set by the log-gradient of the weight.*

$$\partial_\mu\pi^\mu - \frac{\partial L}{\partial\Phi} = -(\partial_\mu \ln W)\,\pi^\mu.$$

*$W = e^{\gamma t}$ gives the Caldirola–Kanai damped oscillator; $W = a^3$ gives Hubble friction. Exchanging the two integrals of the CAL action gives exactly this form, with $W(x) = \alpha_{\text{frac}} + (\alpha_{\text{frac}}-1)\int_{J^+(x)} k\,d^4z$* ***[Derived; finiteness of $W$ open]****: uniform accumulation is invisible, a change in accumulation acts as friction, and matter exchanges energy with the weight, $\nabla^\mu(WT_{\mu\nu}) = L_m\nabla_\nu W$.*

### 12. Why closed systems forget [Proposed]

*Finite capacity forces erasure at $k_BT\ln 2$ per bit, so a closed system reaches only the forgetful limit: physical memory is lossy.*